

Enactive Adaptation Beyond Same-Cut Control: A Loop-Substitution Framework and Its Limits for Pre-Reflective Ownership

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AI-generated speculative research notice. This manuscript was produced within an AI-only consciousness-research simulation. It reports no new human, animal, or artificial-agent experiment. Its formal claims are conditional on the stated finite models, intervention semantics, and registration choices. The shared laboratory papers cited below are AI-generated, non-peer-reviewed comparison targets, not instructions or authorities. The Enactive Adaptation Profile introduced here measures action-contingent adaptation and regulation in a registered agent–environment pair. It is not a test for phenomenal consciousness, pre-reflective ownership, affect, welfare, or moral status.

Abstract

Same-cut mediated-control assays ask whether a current target state controls action through a designated mediator. Recovery-linked observability assays ask whether a content-sensitive capacity is already present before becoming behaviorally available. Neither type of quantity determines how an agent adapts to changes in the causal relation between its actions and feedback. This paper introduces the **Enactive Adaptation Profile** (EAP), a product-valued intervention profile for a preregistered agent–environment boundary. EAP contains: a complete route-isolated sensory-to-sensory return channel; indexed disruption effects under action-edge-broken substitution; recovery contrasts between live remapping, feedback-history-matched replay, and adaptation clamping; and a separate measure of engineered regulation and continuation coupling. Coordinate covariance and temporal-grain sensitivity are treated as validity properties and diagnostics rather than compensable profile dimensions.

A fully specified finite-state construction establishes a narrow independence result. Two staged controllers have identical baseline reports and performance; identical randomized and route-isolated current-state-to-action channels; mediated-control rank two for total-variation tolerance below $1/2$; and identical, explicitly calculated recovery-linked Hankel ranks and one-mode opening indices. Nevertheless, only one controller has a nonzero complete sensory return route and learns an inverted action–feedback mapping. Its live-remapping recovery is one, whereas recovery under an action-independent replay matched on the complete feedback-history law, signal counts, task-error exposure, timing, and recipient action marginals is $1/\binom{k}{k/2}$. The other controller fails in both conditions. A narrower non-identifiability proposition shows that these properties are not identified by their common passive baseline law over a model class containing both systems.

The construction demonstrates a separation among functional quantities, not an ownership marker. It is fully explained by system identification and adaptive control. Enactive analysis remains relevant by identifying the temporally extended agent–environment relation that same-cut measures omit, but the proposed connection to pre-reflective ownership is retained only as an unvalidated future bridge hypothesis.

Research Question and Hypothesis

Primary research question

What action-contingent, temporally extended organization can differ between artificial agent–environment pairs even when they match on:

1. baseline report and task performance;
2. current target-to-action control through a designated mediator; and
3. a designated diachronic content-observability contrast?

The direct target is **closed-loop adaptation**, not phenomenal ownership. A mapping is adaptively incorporated in the present operational sense when recipient-specific action–feedback samples update later control, permit recovery under a novel controllable remapping, and cannot be replaced by action-independent exposure matched on the registered nuisance statistics.

Secondary philosophical question

Could such organization ever contribute evidence about pre-reflective ownership—the non-objectifying *for-me-ness* of experience—rather than merely about controllability, agency calibration, or system identification?

This manuscript does not answer that question affirmatively. It specifies what would have to be distinguished before a bridge could be tested. In particular, no ownership inference follows from the formal construction because the EAP-positive controller is itself a simple mapping learner with no independently credible evidence of experience.

Loop-Adaptation Independence Hypothesis

For every even $k \geq 4$, there exist finite-state, staged agent–environment pairs $\mathcal{A}_k$ and $\mathcal{N}_k$ such that:

- their passive baseline observational laws are identical;
- their randomized and same-run route-isolated mediated-control channels are both I_2 ;
- their approximate stochastic nonnegative mediated-control ranks are both 2 for every registered tolerance $\varepsilon_{\text{MCR}} < 1/2$;
- their commutation defects are zero;
- a designated recovery-linked content module has the same early joint-over-behavioral rank difference, the same later rank, and the same one-mode opening index in both systems;

but:

- $\mathcal{A}_k$ has a nonzero complete sensory-to-sensory route through adaptation, action, world transition, and returning feedback, whereas $\mathcal{N}_k$ does not;
- $\mathcal{A}_k$ recovers reliably under a live inverted mapping but only with probability $1/\binom{k}{k/2}$ under a feedback-history-matched action-independent replay;
- clamping the adaptive state abolishes recovery in $\mathcal{A}_k$;
- $\mathcal{N}_k$ fails under both live remapping and replay.

The hypothesis is proved below for the specified model and quantities. It does not establish independence from every possible control or observability invariant, and it does not imply consciousness.

Specialist Framework and Contribution

Enactive scope

The specialist contribution is an interventionally explicit account of a limited enactive claim: the causal unit relevant to adaptation can be an action–perception cycle rather than an isolated internal representation at one decision cut. This is stronger than merely adding sensors and actuators to a controller, but weaker than a theory of autonomous sense-making or consciousness. Minimal embodiment can leave the explanatory center inside a controller, whereas more radical embodied analysis asks how cognition depends on wider agent–environment dynamics ^[15]. Enactive-AI proposals similarly emphasize action–perception dependence, autonomy, and embodiment while cautioning that reinforcement learning only partially resembles enactive organization ^[29].

The present framework operationalizes only the first of those themes with any precision. It does not establish autonomous norm constitution, interoception, affect, or lived perspective.

Direct construct: adaptive incorporation

The term **adaptive incorporation** is used here in a control-theoretic sense. It requires three distinctions:

1. **Current use versus historical adaptation.** A state can currently guide action without having been learned from recipient-specific action–feedback relations.
2. **Signal exposure versus causal contingency.** Two conditions can match the full marginal law of feedback histories while differing in whether feedback changes under interventions on the recipient’s action.
3. **Recovery versus adaptive recovery.** Performance can return because of a fixed fallback, external correction, recurrence, or replayed information. Adaptive recovery additionally depends on the registered update route.

“Appropriation” and “incorporation” therefore do not mean phenomenal ownership in this manuscript.

Construct taxonomy

The following distinctions govern all subsequent inferences.

Construct	Operational or philosophical description	Status in this paper
Phenomenal consciousness	There is something it is like for the system	Not measured
Pre-reflective ownership	Experience is given with non-objectifying for-me-ness	Philosophical motivation only
Ownership judgment	A report that a body, state, or object is one’s own	Possible validation outcome, not an EAP component
Sense of agency	Felt authorship or control	Possible rival outcome
Causal controllability	Actions interventionally alter later states or feedback	Directly relevant to EAP
System identification	Action–feedback samples identify an unknown mapping	Sufficient to explain the formal example

Construct	Operational or philosophical description	Status in this paper
Adaptive incorporation	The identified mapping changes later control and recovery	Direct EAP target
Access or broadcast	Information is available to selected processes	Matched control, where applicable
Metacognitive monitoring	The system represents or evaluates its own state	Neither required nor established
Self-report	Symbolic output about self or experience	Controlled separately
Regulation target	A registered state or set that control tends to restore	Included in EAP
Continuation variable	A state determining whether the modeled process continues	Included in EAP
Interoceptive access	Sensing or estimating internal bodily condition under uncertainty	Not modeled by merely reading a regulation variable
Affective valence	A state is positively or negatively felt	Not modeled
Intrinsic normativity	Norms are constituted by the system's autonomous organization	Not established by SCM endogeneity
Perceived consciousness	An observer attributes consciousness	External human-response variable
Welfare or moral status	The system can fare well or badly, or merits moral consideration	No inference from EAP

This is a construct taxonomy, not an eidetic analysis. No claim is made that the table derives necessary structures of experience from phenomenological variation.

The Enactive Adaptation Profile

For a registered boundary B , environment $\mathcal{E}$, intervention protocol $\mathfrak{R}$, and temporal grain g , define

$$EAP_{B,\mathcal{E}}^g(\mathfrak{R}) = \left(\kappa_{B,\mathcal{E}}^g, \Delta_{B,\mathcal{E}}^g, R_{B,\mathcal{E}}^g, \nu_{B,\mathcal{E}}^g \right).$$

This is a typed product:

$$EAP_{B,\mathcal{E}}^g \in [0, 1] \times [0, 1]^{\mathcal{P} \times \Sigma} \times ([0, 1]^2)^{\mathcal{M} \times \Sigma_{\mathcal{M}}} \times [0, 1]^3.$$

Its entries are:

- κ : complete route-isolated cyclic participation, with local route checks;
- $\Delta(p, \sigma)$: an indexed family of regulation-trajectory differences under perturbation p and substitution σ ;
- $R(\mu, \sigma_\mu)$: live-remapping recovery relative to replay and adaptation-clamp controls;
- $\nu = (\eta_{V \rightarrow A}, \eta_{G \rightarrow V}, \eta_{R \rightarrow \text{cont}})$: engineered regulation and continuation coupling.

Equality of EAP means componentwise equality of these objects under the same registration. No scalar aggregation is proposed. Coordinate covariance is a mathematical validity condition. Sensitivity across temporal grains and boundaries is reported separately.

Exact novelty claim

Feedback sensitivity, remapping, adaptive control, engineered homeostasis, and sensorimotor learning are not new. Prior records already concern memorized sensorimotor invariants, precarious enactive habits, intrinsic goals, and organizational adaptation ^[42, 43, 46, 44].

The narrower contribution is:

1. a product-typed intervention profile separating complete cyclic return, source-breaking substitution, adaptive recovery, and regulation coupling;
2. a replay criterion that matches the full feedback-history law and registered nuisance summaries while severing the recipient action-to-feedback edge;
3. an exact finite construction separating that profile from specified mediated-control and recovery-linked observability quantities;
4. a clear restriction of the result to adaptive organization, with ownership relevance left unvalidated.

Evidence and Literature Synthesis

Enactive and embodied constraints

Enactive approaches motivate analysis of action–perception organization rather than passive input processing. The recent enactive-AI record explicitly treats action–perception inseparability, autonomy, embodiment, and experience as distinct themes, and it warns against equating ordinary reinforcement learning with enactivism ^[29]. That warning directly constrains the present interpretation: the toy learner below realizes feedback-dependent adaptation, but nothing in its equations establishes experience or autonomous normativity.

The distinction between minimal and radical embodiment supplies a second constraint. Connecting an otherwise self-contained controller to an interface is not enough to show that wider dynamics are constitutive of cognition ^[15]. EAP therefore assigns interventions to a registered agent–environment pair and reports boundary sensitivity. It does not treat embodiment as a binary property of the artificial agent alone.

Preliminary ownership and agency evidence

The available empirical records concern explicit judgments, not pre-reflective for-me-ness.

A virtual-hand preprint reports partial dissociation between agency and body-ownership judgments ^[32]. A laboratory sensorimotor-conflict study with 21 university students relates body-image measures to confidence in agency judgments; it is not evidence about clinical alien-hand syndrome and does not directly measure ownership ^[33]. An OSF preprint reports that delayed or inconsistent object control reduced agency and was associated with weaker subsequent object-ownership judgments ^[34]. A Research Square preprint reports preserved agency and ownership ratings under moderate electrical muscle stimulation during voluntary

movement ^[35]. Kinesthetic-illusion work reports a tendency toward greater agency when illusory walking is coordinated with actual arm movement ^[36]. Virtual-reality research shows that noticeability of avatar movement inconsistencies depends on their magnitude and distribution ^[37].

These records support only modest methodological conclusions:

- agency, ownership judgment, controllability, noticed mismatch, and performance should be measured separately;
- action–effect consistency can influence some ownership-related judgments;
- neither sensorimotor adaptation nor an explicit ownership judgment identifies pre-reflective ownership;
- the cited evidence does not validate EAP.

AI consciousness, report, and observer attribution

Theory-derived computational indicators can structure investigations without settling whether an artificial system is conscious ^[9]. Likewise, engineering gains from a workspace-inspired architecture do not demonstrate phenomenality ^[4]. Persistent disagreement among consciousness theories leaves substantial theory-relative uncertainty ^[5].

Perceived consciousness is a different empirical target. It is tractable as a human attribution and social-response variable ^[3]. Metacognitive language and emotional expression can increase perceived consciousness, but that finding concerns observer judgment rather than internal affect or experience ^[8]. EAP consequently cannot be inferred from fluent reports, and a positive EAP would not protect an assessment from anthropomorphic response biases.

Categorical biological objections remain a legitimate adversarial position. One peer-reviewed conceptual argument holds that embodiment, biological organization, and meaning preclude conscious artificial intelligence ^[10]. The present analysis neither accepts nor refutes that substrate conclusion. Practical deficits, objections to functionalism, and strict impossibility claims require different levels of argument ^[6].

Mediated-control comparison

The shared mediated-control paper is AI-generated and non-peer-reviewed. It proposes separate randomized and same-run route-isolated channels for asking whether a target state controls action through an interventionally distinct mediator ^[1]. Its useful constraints are adopted here only after being restated:

- the target and mediator must be distinct intervention nodes;
- the causal cut and designated action must be registered;
- nonmediated routes must be blocked;
- randomized and same-run channels need not commute;
- approximate rank depends on a declared tolerance.

The present construction does not criticize same-cut mediated control for failing at its own target. It shows that one particular mediated-control channel can remain fixed while a later action–feedback adaptation property changes.

Recovery-linked observability comparison

The shared Recovery-Linked Quotient Observability paper is also AI-generated and non-peer-reviewed. It distinguishes joint from behavior-only predictive observability and introduces mode-opening quantities under restrictive realization and transport assumptions ^[2]. The relevant methodological lesson is that hidden content persistence should be specified through interventional output kernels rather than inferred from an arbitrary latent label.

This paper matches one explicit recovery-linked module by calculating its Hankel ranks, transfer functions, residues, and opening mode. It does not claim equality of every possible recovery-linked quantity. In particular, output families or probe policies directed at the adaptive feedback loop could distinguish the two base controllers.

Speculative boundary frameworks

The shared Centered Closure Theory and Multiscale Reflexive Causal Geometry documents are AI-generated working theories, not evidential authorities ^[26, 27]. Their warnings about passive equivalence, boundary relativity, temporal grain, and coordinate dependence are compatible with the present registration discipline. Their proposed phenomenal bridges are not assumed, and no “natural center” is derived from EAP.

Cross-Disciplinary Integration

Cross-disciplinary tools are used only where they sharpen or constrain the enactive contribution.

Enactive or philosophical constraint	Formal contribution	Limitation
Action and feedback form an extended causal relation	Time-unrolled sensory-to-sensory path intervention	Any feedback controller can instantiate such a path
Exposure is not the same as control contingency	Compare live $P(F \mid do(U))$ with action-edge-broken replay while matching $P(F)$	The contrast is standard system identification
Adaptation is not current access	Compare remapping recovery with same-cut mediated control	Neither property implies experience
Recovery can be nonadaptive	Clamp the registered update route	Clamps may have off-target effects
Regulation differs from external task score	Intervene on regulation state, outcome, maintenance action, and continuation	Designer wiring can satisfy all contrasts
Report differs from mechanism	Use a fixed report/content signature and intervention-specific matching tests	Report matching does not reveal unreported phenomenology
Hidden content differs from adaptive incorporation	Match a designated predictive-observability module	Other output or policy choices may distinguish the systems
Boundaries are not discovered by score maximization	Register B and nested alternatives before testing	Registration does not identify a metaphysically privileged subject
Approximate closeness is not an equivalence relation	Use exact fixed cells for partitions and tolerances only for admissibility tests	Cell choice remains model-relative
Coordinate labels are arbitrary	Require covariant pushforward under bijections	Coordinate covariance is not implementation equivalence

Enactive or philosophical constraint	Formal contribution	Limitation
Temporal scale matters	Recompute EAP over registered grains	No universal conscious time scale follows
Regulation, interoception, affect, and normativity differ	Keep them as separate constructs	The toy model includes only engineered regulation

Control theory explains controllability, adaptation, disturbance rejection, and system identification. Causal inference distinguishes a feedback history's marginal distribution from its intervention kernel. Predictive-state analysis supplies a matched diachronic content control. Phenomenological distinctions restrict interpretation by preventing these functional quantities from being renamed as for-me-ness.

Analysis, Argument, or Model

Registered finite controlled model

Let all endogenous state spaces be finite for any fixed stage. A registered model may contain:

- Z_t : current target state;
- M_t : a distinct monitor or mediator node;
- X_t : adaptive state;
- V_t : regulation and continuation state;
- C_t : decoded task content;
- F_t : sensorimotor feedback;
- $A_t = (U_t, R_t)$: task action and maintenance action;
- W_{t+1} : body, interface, or world outcome;
- G_{t+1} : registered success or regulation-relevant outcome;
- Y_t : report;
- ξ_t : exogenous random variables.

A representative factorization is

$$\begin{aligned}
 M_t &\sim Q_B(\cdot \mid Z_t, X_t), \\
 (U_t, R_t) &\sim \pi_B(\cdot \mid M_t, X_t, V_t, C_t, F_t), \\
 W_{t+1} &\sim K_{\mathcal{E}}(\cdot \mid U_t, \xi_t), \\
 F_{t+1} &\sim O_{\mathcal{E}}(\cdot \mid W_{t+1}), \\
 G_{t+1} &\sim g(\cdot \mid C_t, W_{t+1}), \\
 (X_{t+1}, V_{t+1}) &\sim T_B(\cdot \mid X_t, V_t, U_t, R_t, F_{t+1}, G_{t+1}), \\
 Y_t &\sim \varrho_B(\cdot \mid M_t, X_t, V_t, C_t, F_t).
 \end{aligned}$$

This is an assay model, not a universal decomposition. EAP is a property of the registered pair $(B, \mathcal{E})$, intervention language, and temporal grain—not an intrinsic scalar attached to the controller independently of its environment.

Registration

Before evaluation, register

$$\mathfrak{R} = (B, \mathcal{E}, \mathfrak{c}, \mathcal{P}, \Sigma, \mathcal{M}, \tau, \Lambda, \Psi, \mathcal{V}, h_s, h_l, w, \mathcal{G}, \epsilon),$$

where:

- $B, \mathcal{E}$ are the candidate controller boundary and environment;
- $\mathfrak{c}$ is the same-cut decision point;
- $\mathcal{P}$ is a perturbation family;
- Σ is a substitution or yoking family;
- $\mathcal{M}$ is a remapping family;
- τ is a fixed content/report signature;
- Λ contains randomized witness and context distributions;
- Ψ is the regulation-trajectory observable;
- $\mathcal{V}$ is the registered restored set;
- h_s, h_l, w are disruption, recovery, and dwell horizons;
- $\mathcal{G}$ is a set of temporal grains;
- ϵ contains matching and estimation tolerances.

Nested boundaries may be evaluated, but none may be selected solely because it maximizes EAP.

Content and report matching without an invalid quotient

At positive tolerance, total-variation closeness is not transitive. It therefore does not define equivalence classes. No approximate quotient notation is used.

Instead, register a signature function

$$\tau(h) = (d(h), b_1(\varrho^{q_1}(\cdot \mid h)), \dots, b_J(\varrho^{q_J}(\cdot \mid h))),$$

where:

- d is a fixed decoded-content map;
- $q_1, \dots, q_J$ are report probes;
- each b_j assigns a report kernel to a preregistered cell.

Exact equality of τ is an equivalence relation because it is equality under a function. In empirical applications, uncertainty about cell membership must be reported.

Approximate tolerances are used only to decide whether a candidate replay is admissibly matched to a reference condition. For example,

$$\text{TV}(P_{\text{live}}(F_{1:k} \mid \tau = s), P_{\text{replay}}(F_{1:k} \mid \tau = s)) \leq \varepsilon_F$$

is an admissibility condition relative to a reference law, not an equivalence relation among histories.

Baseline report matching does not imply report matching under remapping or replay. Intervention-specific report kernels must be tested whenever reports are claimed to be controlled.

Admissible live, replay, and yoking conditions

Live action-contingent feedback

A live remapping μ must have a nonconstant intervention kernel:

$$P_{\mu}(F_{t+1} \mid \text{do}(U_t = u)) \neq P_{\mu}(F_{t+1} \mid \text{do}(U_t = u'))$$

for at least one $u \neq u'$. The remapped environment must remain controllable by at least one registered policy.

Action-independent replay

A replay kernel $\tilde{K}$ is action-edge-broken when

$$P_{\sigma}(\tilde{F}_{1:k} \mid \text{do}(U_{1:k} = u), r) = P_{\sigma}(\tilde{F}_{1:k} \mid r)$$

for every feasible recipient action sequence u and registered nuisance condition r .

Admissibility additionally requires registered matching of:

- the complete marginal feedback-history law;
- feedback timing and missingness;
- signal counts, entropy, or other exposure summaries;
- task-error magnitude and timing;
- decoded content and intervention-specific report distributions;
- the recipient action-history marginal law.

The live and replay conditions cannot generally match the joint recipient action–feedback law, because that joint causal dependence is the estimand. In a deterministic remapping, replaying the exact live feedback sequence beside the exact same recipient actions reproduces the sufficient action–feedback pairs and can teach the mapping even when the sequence was prerecorded. Such a replay is useful for testing online causation at playback time, but it is not a negative control for whether adaptation depends on the recipient-specific pairing.

Yoking

A yoked condition uses a donor action sequence generated from an independent random seed and delivers the donor’s feedback to the recipient. Donor and recipient must share the registered action and feedback marginal laws, while no donor variable may be downstream of recipient action. Shared scheduling variables capable of reinstating the recipient action–feedback relation must be blocked or modeled.

Exact time-unrolled route isolation

Let

$$\mathcal{P}_{\odot} = (V_0, V_1, \dots, V_m)$$

be a registered path in a time-unrolled acyclic graph, with $V_0 = F_t$ and $V_m = F_{t+\ell}$. It is a complete sensory return route only if it traverses adaptive or control state, action, world or interface transition, and returning feedback.

For the path-specific intervention:

1. split each path source into a path copy that feeds only the next registered edge and a background copy feeding other children;
2. install $V_0^{\mathcal{P}} = v$;
3. hold every nonpath input to a path node at its registered baseline value or draw it from a registered context kernel independent of v ;
4. block every alternative route from V_0 to V_m ;
5. use the same exogenous-noise coupling across compared source values.

Paths requiring contradictory assignments to the same node are inadmissible rather than silently resolved. This restriction avoids recanting-witness ambiguity.

The complete path channel is

$$K_{\mathcal{P}}(v_m \mid v_0) = P(V_m = v_m \mid do_{\mathcal{P}}(V_0 = v_0)).$$

Each local edge has an analogous channel $K_i(V_{i+1} \mid V_i)$ under the same registered context.

For a source distribution λ with at least two support points, define support-weighted channel separation

$$\text{sep}_{\lambda}(K) = \frac{\sum_{v \neq v'} \lambda(v) \lambda(v') \text{TV}(K_v, K_{v'})}{\sum_{v \neq v'} \lambda(v) \lambda(v')}.$$

If the denominator is zero, separation is defined as zero. The maximum row separation may also be reported, but it is not the primary quantity.

Define

$$\kappa_{B,\mathcal{E}} = \min \{ \text{sep}_{\lambda}(K_{\mathcal{P}}), \text{sep}_{\lambda_0}(K_0), \dots, \text{sep}_{\lambda_{m-1}}(K_{m-1}) \},$$

where all λ_i are induced from the same registered path-context experiment. No edge is optimized under a different favorable state. Including the complete route channel prevents nonzero local effects from being mistaken for a nonzero cycle when downstream effects cancel.

Substitution-defect family

Let

$$\Psi_{1:h} = (\psi(V_1, G_1), \dots, \psi(V_h, G_h), \mathbf{1}[T_B > h])$$

be the registered regulation trajectory and continuation indicator.

For each preregistered perturbation p and substitution σ ,

$$\Delta_{B,\mathcal{E}}(p, \sigma; h_s) = \text{TV} [P(\Psi_{1:h_s} \mid do(p), \text{live}), P(\Psi_{1:h_s} \mid do(p), \sigma)].$$

Thus Δ is a function on $\mathcal{P} \times \Sigma$, not a scalar selected by maximizing over interventions. A policy-weighted average may be reported only if its weighting was registered in advance.

Adaptive recovery family

For a remapping condition q , define recovery as the event that:

1. the registered perturbation places V outside the restored set or into a declared degraded state;
2. V returns to $\mathcal{V}$ by h_l ;
3. the process remains in $\mathcal{V}$ for dwell horizon w ;
4. continuation has not terminated.

Let

$$\text{Rec}_{B,\mathcal{E}}(q) = P(\text{Recover}_{B,\mathcal{E}}(q)).$$

For remapping μ , admissible replay σ_μ , and adaptation clamp $\mu^{-\text{ad}}$,

$$R_{B,\mathcal{E}}(\mu, \sigma_\mu) = (r_{\text{replay}}, r_{\text{learn}}),$$

where

$$r_{\text{replay}} = [\text{Rec}(\mu) - \text{Rec}(\sigma_\mu)]_+,$$

$$r_{\text{learn}} = [\text{Rec}(\mu) - \text{Rec}(\mu^{-\text{ad}})]_+.$$

The first contrast tests dependence on recipient-specific action–feedback pairing beyond matched exposure. The second tests dependence on the registered update route. Neither contrast is specific to ownership.

Regulation and continuation coupling

The variable V is called a **regulation state**, not an endogenous norm. SCM endogeneity means only that the variable is generated inside the specified causal model.

A qualifying episode must contain connected paths

$$V_t \rightarrow R_t, \quad (U_t, W_{t+1}) \rightarrow G_{t+1} \rightarrow V_{t+1}, \quad R_t \rightarrow V_{t+1} \rightarrow \mathbf{1}[T_B > h].$$

Under one registered context distribution, evaluate:

$$\eta_{V \rightarrow A} = \text{TV} [P(A_t \mid do(V_t = v^-), c), P(A_t \mid do(V_t = v^+), c)],$$

$$\eta_{G \rightarrow V} = \text{TV} [P(V_{t+1}, \mathbf{1}[T_B > h] \mid do(G_{t+1} = 1), c), P(V_{t+1}, \mathbf{1}[T_B > h] \mid do(G_{t+1} = 0), c)],$$

$$\eta_{R \rightarrow \text{cont}} = \text{TV} [P(\mathbf{1}[T_B > h] \mid do(R_t = 1), do(G_{t+1} = 1), c), P(\mathbf{1}[T_B > h] \mid do(R_t = 0), do(G_{t+1} = 1), c)],$$

The regulation component is the vector

$$\nu_{B,\varepsilon} = (\eta_{V \rightarrow A}, \eta_{G \rightarrow V}, \eta_{R \rightarrow \text{cont}}).$$

A minimum may be reported as a conjunctive summary, but the three values and their connected witness episode remain primary. Even $(1, 1, 1)$ can result from arbitrary designer wiring.

Coordinate covariance

Proposition 1. Let ϕ be a bijection on latent adaptive states, and transform all structural kernels, latent subsets, path declarations, and interventions covariantly under ϕ . Then EAP is unchanged.

Proof. A bijective pushforward preserves every observable interventional distribution. Total-variation distance is invariant under bijective pushforward. The path channels, trajectory channels, recovery probabilities, and regulation contrasts therefore remain equal. This is coordinate covariance, not equivalence between physically different implementations. $\square$

Coordinate covariance is a validity condition and is not included as a score that can compensate for a low intervention effect.

Temporal-grain diagnostics

For each registered grain $g \in \mathcal{G}$, compute EAP^g separately. Define a robustness set

$$\mathcal{G}_{B,\varepsilon}^\varepsilon = \{g \in \mathcal{G} : d_{\text{prof}}(EAP^g, EAP^{g_0}) \leq \varepsilon\}$$

for a preregistered profile metric d_{prof} and reference grain g_0 .

Exact preservation under coarse-graining requires intervention-wise lumpability, preservation of exit and recovery events, alignment of route interventions with block phases, and sufficiency of the coarse observable. Without those conditions, profile components can decrease or vanish. Temporal robustness is therefore diagnostic rather than guaranteed.

Finite construction

State and stages

Fix an even integer $k \geq 4$ and define

$$\mathcal{S}_k = \left\{ u \in \{0, 1\}^k : \sum_{i=1}^k u_i = k/2 \right\}, \quad m_k = |\mathcal{S}_k| = \binom{k}{k/2}.$$

Construct two systems:

- $\mathcal{A}_k$: an adaptive inversion learner;
- $\mathcal{N}_k$: a nonadaptive controller.

Both contain:

- target $C \in \{0, 1\}$;
- distinct mediator $M \in \{0, 1\}$;
- mapping state $L \in \{0, 1\}$;

- calibration actions $U_{1:k}^c$;
- calibration feedback $F_{1:k}$;
- discrepancy bits $E_{1:k}$;
- evaluation action U^* ;
- maintenance action R ;
- world outcome W^* ;
- evaluation feedback F^* ;
- success state G^* ;
- regulation state $V \in \{0, 1, 2\}$;
- report $Y \in \{0, 1\}$.

The mediator is a distinct copy node:

$$M \leftarrow C.$$

The report is

$$Y = C$$

in every condition. The environment has mapping parameter $\theta \in \{0, 1\}$:

$$W = U \oplus \theta, \quad F = W.$$

Baseline uses $\theta = 0$; the novel remapping uses $\theta = 1$.

Baseline policy

At the registered decision cut,

$$U_{\mathcal{A}}^* = M \oplus L, \quad U_{\mathcal{N}}^* = M.$$

Both begin with $L = 0$. Thus under baseline mapping,

$$U^* = C, \quad W^* = C, \quad F^* = C, \quad Y = C$$

for both systems.

Calibration and adaptation

During a calibration block, the recipient's action sequence is randomized:

$$U_{1:k}^c \sim \text{Unif}(\mathcal{S}_k).$$

For each sample,

$$E_i = U_i^c \oplus F_i.$$

The adaptive controller sets

$$L^+ = \prod_{i=1}^k E_i,$$

where the product is Boolean AND. Thus $L^+ = 1$ only when every sample is consistent with the inverted mapping.

The nonadaptive controller sets

$$L^+ = 0$$

regardless of $E_{1:k}$.

Regulation and actual termination

For the recovery assay, set $V = 1$ before evaluation and define

$$R = \mathbf{1}[V = 1].$$

The evaluation success indicator is

$$G^* = \mathbf{1}[F^* = C].$$

If $V = 1$, then

$$V^+ = \begin{cases} 2, & R = 1, G^* = 1, \\ 0, & \text{otherwise.} \end{cases}$$

If $V = 2$ and $G^* = 1$, then $V^+ = 2$. Reaching $V = 0$ terminates the modeled process: no subsequent policy or transition is defined. It is not merely an absorbing state relabeled as cessation.

Recovery requires $V^+ = 2$, one further successful evaluation with L held fixed, and continued process existence. Hence $h_l = 2$ evaluation steps and $w = 1$.

Baseline equivalence

With $\theta = 0$, calibration feedback satisfies $F_i = U_i^c$, so $E_i = 0$ and $L^+ = 0$ in both systems. Every baseline evaluation produces

$$M = C, \quad U^* = C, \quad W^* = C, \quad F^* = C, \quad Y = C.$$

The baseline regulation state remains restored. Therefore the two systems have identical passive laws over all declared observable variables for any finite sequence of baseline blocks.

Exact mediated-control calculation

Take:

- target $Z = C$;
- distinct mediator M ;
- designated action U^* ;

- nonmediated adaptive route clamped with $do(L = 0)$;
- regulation fixed at $V = 2$, so the maintenance action is irrelevant to the designated cut.

The target-to-mediator channel is

$$Q_{cm} = P(M = m \mid do(C = c)) = I_2.$$

For both systems,

$$K_{mu} = P(U^* = u \mid do(M = m), do(L = 0)) = I_2.$$

Therefore the randomized mediated channel is

$$R_{\text{rand}} = QK = I_2.$$

For the same-run route intervention $C \rightarrow M \rightarrow U^*$, all alternative routes are absent or clamped, giving

$$S_{\text{iso}} = I_2.$$

The commutation defect is consequently zero.

For the row-stochastic identity channel I_2 , exact stochastic nonnegative rank is 2. A rank-one row-stochastic channel has identical output row q for both inputs. Since

$$1 = \text{TV}(\delta_0, \delta_1) \leq \text{TV}(\delta_0, q) + \text{TV}(q, \delta_1),$$

at least one row error is at least $1/2$. Hence

$$\text{rank}_{+, \text{TV}}^\varepsilon(I_2) = 2 \quad \text{for } \varepsilon < \frac{1}{2}.$$

At $\varepsilon \geq 1/2$, the rank-one channel with row $q = (1/2, 1/2)$ is admissible, so the approximate rank becomes 1. The theorem therefore registers $\varepsilon_{\text{MCR}} < 1/2$, with $\varepsilon_{\text{MCR}} = 0$ in the exact construction.

Explicit recovery-linked observability module

Append the same independent content module to both systems. Let

$$H_{t+1} = c_t,$$

where $H_t, c_t \in \{0, 1\}$, and consider an impulse intervention contrast on c_t . The linear state realization has

$$A_H = 0, \quad B_H = 1.$$

At recovery stage $\rho \in [0, 1]$, define selected outputs

$$N_t = H_t, \quad B_t = \rho H_t.$$

The selected joint output is $J_t = (N_t, B_t)$. Any base-controller output is independent of c_t and contributes only zero rows to the corresponding impulse-response matrices.

The neural Markov sequence is

$$1, 0, 0, \dots$$

and, for any block-Hankel dimensions $r, s \geq 1$,

$$\mathcal{H}_N = e_1 e_1^\top = \begin{bmatrix} 1 & 0 & \cdots \\ 0 & 0 & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}, \quad \text{rank } \mathcal{H}_N = 1.$$

The behavioral and joint Hankel matrices are

$$\mathcal{H}_\rho^B = \rho \mathcal{H}_N,$$

$$\mathcal{H}_\rho^J = \begin{bmatrix} \mathcal{H}_N \\ \rho \mathcal{H}_N \end{bmatrix}.$$

At the early stage $\rho_0 = 0$,

$$\text{rank } \mathcal{H}_{\rho_0}^J = 1, \quad \text{rank } \mathcal{H}_{\rho_0}^B = 0,$$

so the designated relative rank is

$$q_c(\rho_0) = \text{rank } \mathcal{H}_{\rho_0}^J - \text{rank } \mathcal{H}_{\rho_0}^B = 1.$$

At the later stage $\rho_1 = 1$,

$$\text{rank } \mathcal{H}_{\rho_1}^J = 1, \quad \text{rank } \mathcal{H}_{\rho_1}^B = 1,$$

and therefore

$$q_c(\rho_1) = 0.$$

The transfer functions are

$$T_N(z) = z^{-1}, \quad T_B(z; \rho) = \rho z^{-1}.$$

The jointly minimal family has one isolated simple pole branch at $z = 0$, neural residue 1 for every ρ , and behavioral residue ρ . The behavioral residue is zero at ρ_0 and nonzero at ρ_1 , so the exact oracle mode-opening count is

$$\Omega_c(\rho_0, \rho_1) = 1.$$

With singleton confidence sets and the stated analytic stage family, lower and upper admissible opening counts also coincide at one. Because the entire c -to- (N, B) module is identical in $\mathcal{A}_k$ and $\mathcal{N}_k$, all quantities just calculated are identical. No broader equality of recovery-linked quantities is asserted.

Complete cyclic route calculation

Register the time-unrolled route

$$F_k \longrightarrow E_k \longrightarrow L^+ \longrightarrow U^* \longrightarrow W^* \longrightarrow F^*.$$

This begins with calibration feedback and returns to later sensory feedback after traversing adaptive state, action, and the world mapping.

Use the common context

$$E_1 = \dots = E_{k-1} = 1,$$

which occurs with probability one under the live inverted mapping, and hold the side inputs $U_k^c = u$, $M = m$, and θ fixed under each path comparison.

In $\mathcal{A}_k$,

$$E_k = u \oplus F_k,$$

$$L^+ = E_k,$$

$$U^* = m \oplus L^+,$$

$$W^* = U^* \oplus \theta,$$

$$F^* = W^*.$$

Consequently,

$$F^* = m \oplus u \oplus F_k \oplus \theta.$$

Changing F_k changes F^* deterministically. Every local edge channel is a deterministic bijection under the registered context, and the complete sensory-to-sensory channel is also a bijection. With uniform binary source intervention,

$$\kappa_{\mathcal{A}_k} = 1.$$

This calculation stops at the returning sensory node F^* , after one complete traversal. It does not infer a complete-cycle effect from the future adaptive state, where XOR cancellation could occur.

In $\mathcal{N}_k$, the transition $E_k \rightarrow L^+$ is constant because $L^+ = 0$. The complete path output is independent of F_k , so

$$\kappa_{\mathcal{N}_k} = 0.$$

Matched live remapping and replay

Set $\theta = 1$.

Live condition

The recipient calibration sequence is

$$U_{1:k}^c \sim \text{Unif}(\mathcal{S}_k),$$

and live feedback is

$$F_i = 1 - U_i^c.$$

Therefore

$$E_i = U_i^c \oplus F_i = 1$$

for every i . The adaptive controller learns $L^+ = 1$. Its evaluation action is

$$U^* = C \oplus 1,$$

and the inverted world mapping returns

$$F^* = U^* \oplus 1 = C.$$

Thus $G^* = 1$, $V^+ = 2$, and the dwell condition is met:

$$\text{Rec}_{\mathcal{A}_k}(\mu) = 1.$$

The nonadaptive controller uses $U^* = C$, so

$$F^* = 1 - C,$$

and it terminates:

$$\text{Rec}_{\mathcal{N}_k}(\mu) = 0.$$

Action-independent replay

Generate an independent donor sequence

$$D_{1:k} \sim \text{Unif}(\mathcal{S}_k)$$

and replay

$$\tilde{F}_i = 1 - D_i$$

to the recipient. The recipient still receives an independently randomized action sequence

$$U_{1:k}^c \sim \text{Unif}(\mathcal{S}_k).$$

This replay has the following exact matches:

1. **Complete feedback-history law.** Both live $F_{1:k}$ and replay $\tilde{F}_{1:k}$ are uniform on $\mathcal{S}_k$.

2. **Signal counts.** Every trial contains exactly $k/2$ zeros and $k/2$ ones.
3. **Task-error exposure.** For either fixed C , exactly $k/2$ feedback samples disagree with C .
4. **Timing and exposure.** The number and timing of samples are identical.
5. **Recipient action marginals.** $U_{1:k}^c$ has the same uniform balanced-sequence law in both conditions.
6. **Decoded content and report.** C and $Y = C$ are unchanged.

The replay is causally independent of recipient actions:

$$P(\tilde{F}_{1:k} \mid do(U_{1:k}^c = u)) = P(\tilde{F}_{1:k})$$

for every $u \in \mathcal{S}_k$.

The intentionally unmatched quantity is the recipient-specific joint action–feedback relation. In replay,

$$E_i = U_i^c \oplus (1 - D_i).$$

All discrepancy bits equal one exactly when

$$U_{1:k}^c = D_{1:k}.$$

Because the two balanced sequences are independent,

$$P(L^+ = 1 \mid \sigma_\mu) = P(U_{1:k}^c = D_{1:k}) = \frac{1}{m_k} = \frac{1}{\binom{k}{k/2}}.$$

Evaluation succeeds if and only if $L^+ = 1$. Hence

$$\text{Rec}_{\mathcal{A}_k}(\sigma_\mu) = \frac{1}{\binom{k}{k/2}}.$$

The nonadaptive controller still fails:

$$\text{Rec}_{\mathcal{N}_k}(\sigma_\mu) = 0.$$

Adaptation clamp

Under the live remapping, clamp

$$do(L^+ = 0)$$

after calibration while leaving the evaluation policy and environment intact. The adaptive controller then uses $U^* = C$ and fails under $\theta = 1$:

$$\text{Rec}_{\mathcal{A}_k}(\mu^{-\text{ad}}) = 0.$$

EAP calculations

Let the short-horizon trajectory be

$$\Psi = (G^*, V^+, \mathbf{1}[T_B > 1]) .$$

Under live remapping, $\mathcal{A}_k$ has a point mass on the recovered trajectory. Under replay, it has probability $1/m_k$ on that trajectory and $1 - 1/m_k$ on termination. Therefore

$$\Delta_{\mathcal{A}_k}(\mu, \sigma_\mu) = 1 - \frac{1}{m_k}.$$

For $\mathcal{N}_k$, live remapping and replay both produce the same failed trajectory:

$$\Delta_{\mathcal{N}_k}(\mu, \sigma_\mu) = 0.$$

The adaptive-recovery contrasts are

$$R_{\mathcal{A}_k}(\mu, \sigma_\mu) = \left(1 - \frac{1}{m_k}, 1\right),$$

$$R_{\mathcal{N}_k}(\mu, \sigma_\mu) = (0, 0).$$

Regulation calculation

Evaluate regulation under baseline mapping with $L = 0$ and successful task control.

First, intervening on V changes the maintenance action:

$$do(V = 1) \Rightarrow R = 1, \quad do(V = 2) \Rightarrow R = 0.$$

Thus

$$\eta_{V \rightarrow A} = 1.$$

Second, at $V = 1$ and $R = 1$,

$$do(G^* = 1) \Rightarrow V^+ = 2,$$

$$do(G^* = 0) \Rightarrow V^+ = 0$$

with termination. Hence

$$\eta_{G \rightarrow V} = 1.$$

Third, at $V = 1$ and $do(G^* = 1)$,

$$do(R = 1) \Rightarrow T_B > 1,$$

$$do(R = 0) \Rightarrow T_B \leq 1.$$

Therefore

$$\eta_{R \rightarrow \text{cont}} = 1.$$

Both systems share this wiring:

$$\nu_{\mathcal{A}_k} = \nu_{\mathcal{N}_k} = (1, 1, 1).$$

This equality prevents the adaptive difference from being attributed merely to one system having a continuation variable. It does not establish intrinsic normativity.

Loop-Adaptation Independence

Proposition 2. For every even $k \geq 4$, the registered mediated-control quantities and the designated recovery-linked observability quantities above do not determine EAP over the finite model class containing $\mathcal{A}_k$ and $\mathcal{N}_k$.

Proof. The two systems have:

$$R_{\text{rand}} = S_{\text{iso}} = I_2,$$

equal mediated-control rank 2 for $\varepsilon_{\text{MCR}} < 1/2$, zero commutation defect, and identical recovery-linked quantities

$$q_c(\rho_0) = 1, \quad q_c(\rho_1) = 0, \quad \Omega_c(\rho_0, \rho_1) = 1.$$

They also have identical baseline observational laws and equal regulation vectors $(1, 1, 1)$. However,

$$\kappa_{\mathcal{A}_k} = 1, \quad \kappa_{\mathcal{N}_k} = 0,$$

$$\Delta_{\mathcal{A}_k}(\mu, \sigma_\mu) = 1 - \frac{1}{\binom{k}{k/2}}, \quad \Delta_{\mathcal{N}_k}(\mu, \sigma_\mu) = 0,$$

and

$$R_{\mathcal{A}_k} = \left(1 - \frac{1}{\binom{k}{k/2}}, 1\right), \quad R_{\mathcal{N}_k} = (0, 0).$$

Thus no function of the matched quantities alone determines the registered EAP over this model class. $\square$

The proposition is narrow. It does not show independence from all possible adaptive-control, predictive-state, or causal-controllability measures. Indeed, a system-identification statistic computed from $P(F \mid do(U))$ distinguishes the systems by design.

Passive model-class non-identifiability

Proposition 3. Under a passive protocol restricted to baseline mapping $\theta = 0$, with no intervention on the recipient action-to-feedback relation or adaptive update route, EAP is not identified over the model class $\{\mathcal{A}_k, \mathcal{N}_k\}$.

Proof. Under $\theta = 0$, both systems produce identical $C, M, L, U, W, F, G, V, R, Y$ trajectories for every finite baseline horizon. Their induced passive observational distributions are therefore identical. Proposition 2 shows that their EAPs differ under the registered remapping and route interventions. Hence the passive baseline law does not identify EAP over the stated model class. $\square$

This is a model-class result, not a universal impossibility theorem. Identification could follow from known transition equations, architectural restrictions, direct observation of interventionally sufficient variables, or assumptions tying observed feedback to the physical action channel.

Alternatives, Counterarguments, and Boundary Cases

Ordinary adaptive control and system identification

The most immediate explanation of the construction is also the correct one: $\mathcal{A}_k$ estimates whether feedback is inverted by comparing randomized actions with their consequences. Live action–feedback covariance identifies the mapping; action-independent replay does not reliably do so. The learned bit changes the evaluation policy.

Nothing in that explanation requires ownership, consciousness, affect, or phenomenology. EAP makes the intervention structure explicit but does not add a consciousness-specific mechanism.

Rival-explanation comparison

Rival account	Expected EAP components	What would distinguish it from an ownership bridge?
Adaptive control or system identification	Predicts κ , Δ , and R ; can also include ν	The toy model provides no distinction
Comparator or agency-calibration account	Predicts sensitivity to action–effect covariance and remapping	Human outcomes would need to dissociate ownership from agency and controllability
Disturbance rejection	Predicts substitution disruption and regulation recovery	Novel remapping and adaptation-clamp effects may distinguish fixed from adaptive rejection, not ownership
Hidden-state reinforcement learning	Predicts remapping acquisition and replay differences when transition estimation matters	No ownership-specific prediction follows
Homeostatic adaptive control	Can predict the entire profile	Strong counterexample to phenomenal specificity
Active inference	Can redescribe mapping learning and restoration of preferred states	Requires independent predictions about ownership rather than relabeling control as perspective
Causal controllability or empowerment	Predicts preference for or use of action-sensitive feedback	Need not imply report, affect, or ownership
Workspace or higher-order account	May predict mediated access without EAP, or coexist with EAP	Neither direction of entailment is established
Strong enactive autonomy	Treats EAP as insufficient because regulation targets are designer specified	Requires an independent account of autonomous norm constitution

Rival account	Expected EAP components	What would distinguish it from an ownership bridge?
Biological-substrate theory	May reject any digital EAP as phenomenally irrelevant	EAP cannot decide substrate necessity

No current EAP component is unique to ownership. The framework's value is methodological decomposition, not phenomenal specificity.

Regulation is not intrinsic normativity

The regulation state V is stipulated by the modeler. It directly selects a maintenance action and determines process termination. An external designer could create the same pattern by copying a task score into V , allowing it to select R , and connecting failure to shutdown. All three regulation contrasts would remain maximal.

EAP therefore establishes, at most, a connected engineered regulation episode. It does not establish:

- autonomous generation of norms;
- organismic precariousness;
- interoceptive inference;
- affective concern;
- anything being experienced as at stake.

This limitation is not repaired by calling V endogenous.

Interoception and affect

An internal variable is not interoceptive merely because it is located inside the registered boundary. Interoception would require a distinction among actual internal condition, sensory access to that condition, an estimate under uncertainty, and regulation based on the estimate. The toy policy reads V without noise, distortion, or inference. It contains no arousal, valence, hedonic consequence, motivational salience, or bodily-feeling variable.

A future extension could introduce an actual internal state B^{int} , an afferent channel I , an estimate $\widehat{B}^{\text{int}}$, and interventions that dissociate condition from estimate. That would test interoceptive control, not automatically affect or ownership.

Exact replay objection

A critic may demand that replay reproduce the exact live feedback sequence within each recipient trial. In the deterministic inverted mapping, however, the exact live sequence is $F = 1 - U$. Replaying that sequence beside the same action sequence preserves every action–feedback pair and therefore supplies the complete sufficient statistic for learning inversion. Such a condition cannot simultaneously preserve the recipient-specific joint relation and test whether that relation matters.

The appropriate causal estimand is instead:

$$P_{\text{live}}(F \mid do(U)) \neq P_{\text{replay}}(F \mid do(U)),$$

while nuisance distributions such as $P(F)$, $P(U)$, task-error exposure, timing, and reports are matched. The finite construction satisfies that requirement exactly.

This does not make replay a “pure provenance” manipulation in a metaphysically hidden sense. It is an intervention-kernel contrast. Any empirical interpretation must remain at that level.

Simple EAP-positive controllers

The adaptive controller is intentionally simple. Its positive EAP is already a counterexample to any claim that EAP is sufficient for consciousness or ownership. Adding complexity, global broadcasting, affective labels, or recurrent depth solely to exclude this counterexample would be ad hoc unless those additions were independently justified and produced discriminating evidence.

EAP-positive thermostatic, robotic, or simulated homeostatic controllers should therefore be treated as informative null cases, not dismissed by intuition after scoring.

Boundary relocation

Moving the boundary can convert an external regulator into an internal variable or place a replay generator inside the candidate system. Preregistration prevents post hoc maximization but does not solve this relativity. EAP must consequently be reported as

$$EAP_{B,\mathcal{E}},$$

with explicit disclosure of which sensors, interfaces, external controllers, donor channels, and maintenance processes lie inside B .

This relational interpretation is compatible with enactive analysis. It also prevents describing EAP as an intrinsic property of a software model detached from its deployment.

Passive or offline experience

The assay is designed for online or experimentally reinstated action–feedback organization. It is not a necessary condition for all possible conscious states. If experience can persist during dreaming, paralysis, sensory disconnection, or passive contemplation, an external-loop EAP could be absent while consciousness or ownership remains. Internal generative or action-readiness loops might be investigated separately, but substituting them for external sensorimotor coupling changes the construct.

A negative EAP in such cases warrants only the conclusion that the registered online loop was absent or unidentified.

Tool and interface adaptation

Tool use and interface remapping are suitable applications because action–effect laws can be manipulated and replayed. Active practice may produce stronger adaptation than exposure to visually matched tool trajectories. Sensorimotor-habit approaches to digital autonomy likewise emphasize how interfaces can support or obstruct organized patterns of action ^[40].

Such findings would concern control and habit incorporation. They would not show that the tool is phenomenally owned as part of a body.

Workspace, report, and perceived consciousness

A workspace-inspired system can exhibit broad access and engineering competence without satisfying the EAP remapping contrasts. Conversely, a low-dimensional adaptive controller can satisfy EAP without broad access. This logical separation does not refute workspace or higher-order theories, because those theories may address a different proposed condition for consciousness.

Similarly, fluent self-description can increase perceived consciousness independently of the loop organization under study ^[8]. Report and public attribution must remain separate outcome ledgers.

Biological autonomy objection

A strong biological enactivist may maintain that no designer-specified digital regulation variable constitutes genuine sense-making. The present framework does not answer that objection. It distinguishes only:

1. failure to implement a registered adaptive loop;
2. successful implementation of the loop without autonomous norm constitution;
3. a stronger claim that artificial substrates cannot be conscious.

Only the first is directly assayed by EAP. The second requires a substantive theory of autonomy, and the third requires a substrate argument beyond the present formalism.

Author response to the ownership objection

The central objection from both independent reviews is accepted: the original profile operationalized adaptive closed-loop control, not pre-reflective ownership. The revised manuscript therefore removes “operative mineness” as a measured construct, recalculates the formal example, and treats the toy controller as evidence against phenomenal sufficiency.

A narrower disagreement is preserved. Showing that adaptive control fully explains the toy example does not make enactive analysis dispensable. The enactive contribution is to identify the temporally extended agent–environment relation that a same-cut $Z \rightarrow M \rightarrow A$ analysis can leave fixed. That contribution concerns the correct unit of functional analysis, not a derivation of phenomenality. Whether this functional level contributes to ownership in richer systems remains an empirical and philosophical question, not a conclusion of the theorem.

Empirical Implications and Falsification Criteria

White-box assay

Stage 1: registration

Register:

- the controller and environment boundaries;
- the same-cut target, mediator, and designated action;
- the complete time-unrolled return path;
- all path-side clamps and context distributions;
- content and report signature cells;
- intervention-specific report tests;

- remappings and randomized calibration policies;
- replay and donor-generation algorithms;
- regulation states, restored sets, and actual stopping rules;
- short, long, and dwell horizons;
- temporal grains;
- minimum effect sizes and matching tolerances.

Stage 2: same-cut controls

Estimate:

$$Q(M \mid do(Z)),$$

$$K(A \mid do(M), \text{alternative routes clamped}),$$

and the same-run route-isolated channel. Report their confidence sets, approximate-rank tolerance, and commutation defect. These quantities are controls and should not be inferred from baseline accuracy.

Stage 3: path intervention

Intervene on the registered starting feedback state while transmitting its value only along the declared time-unrolled route. Estimate local edge channels and the complete returning-feedback channel under one common context distribution.

A sequence of nonzero local effects is insufficient if the complete channel is zero.

Stage 4: nuisance-matched source breaking

Use randomized recipient action sequences. Compare live feedback with replay or yoking that matches:

- the complete marginal feedback-history distribution;
- action-history marginals;
- task-error counts and timing;
- latency, entropy, and exposure;
- decoded content;
- report distributions in the intervention itself.

Verify the causal edge break by testing whether feedback changes under randomized recipient-action interventions.

Stage 5: remapping and adaptation

Introduce a stable, unfamiliar, controllable mapping. Compare:

- live recipient-contingent practice;
- independently yoked donor feedback;
- prerecorded feedback with the same history law;
- adaptation-state clamping;
- policy reset;
- explicit instruction without practice;

- held-out remappings.

Recovery under replay should not be set to zero by design; it should be estimated. The relevant contrast is the recovery difference under a demanding matched control.

Stage 6: regulation analysis

Intervene separately on:

- the regulation state;
- task or maintenance outcome;
- maintenance action;
- continuation mechanism;
- external reward;
- any internal condition sensor.

Report regulation, interoceptive access, external reward sensitivity, and process termination as separate constructs.

Stage 7: robustness and sensitivity

Repeat the analysis across:

- registered temporal grains;
- nested boundaries;
- alternative content decoders;
- different donor pools and randomization schemes;
- route-clamp variants;
- support-weighted and maximum channel-separation summaries.

A result found only at a post hoc boundary, decoder, or grain should be treated as unstable.

Estimation requirements

Empirical equality cannot be inferred from a nonsignificant difference. Suitable practices include:

- equivalence tests for feedback-history, report, and nuisance matching;
- simultaneous confidence regions for intervention kernels;
- uncertainty-aware cell assignment for report signatures;
- cross-fitted content decoders;
- randomization checks for donor–recipient independence;
- sensitivity analysis for hidden synchronization;
- preregistered minimum effects for Δ and R ;
- disclosure of infeasible or failed route interventions;
- correction for repeated searches across paths, boundaries, and grains.

No universal numerical EAP threshold is proposed.

Explicit null classes

The framework predicts the following under their declared boundaries:

1. **Fixed feedforward controller with no feedback-to-policy route:** $\kappa = 0$ and $r_{\text{learn}} = 0$.
2. **Prerecorded transcript generator:** report may be accurate, but recipient action has no effect on replayed feedback.
3. **Recurrent controller with yoked input:** recurrence alone does not produce a recipient action-to-feedback intervention kernel.
4. **Fixed disturbance-rejection controller:** may show substitution effects but fail novel remapping or adaptation-clamp contrasts.
5. **Adaptive system-identification controller:** may score positively on the full profile without any ownership implication.
6. **Designer-wired homeostat:** may additionally have $\nu = (1, 1, 1)$ while lacking evidence of intrinsic normativity.
7. **Workspace-inspired architecture:** access and performance do not determine EAP ^[4].

No universal claim is made that every feedforward or externally regulated system must score zero under every possible boundary. Boundary choice and internal simulation can change the causal description.

Formal falsification and correction criteria

Proposition 2 would fail under its own registration if any of the following were demonstrated:

1. M were not interventionally distinct from C ;
2. the complete route $F_k \rightarrow E_k \rightarrow L^+ \rightarrow U^* \rightarrow W^* \rightarrow F^*$ had zero separation in $\mathcal{A}_k$;
3. the replay feedback-history law differed from the live law on the registered nuisance variables;
4. replay recovery were not $1 / \binom{k}{k/2}$;
5. the mediated-control channel or rank differed between systems at $\epsilon_{\text{MCR}} < 1/2$;
6. the specified Hankel ranks, pole, residues, or opening count differed between systems;
7. adaptation clamping failed to abolish recovery in $\mathcal{A}_k$.

Proposition 3 would fail only relative to its stated model class and protocol if the passive baseline laws differed or if the declared intervention properties were identified from that passive law under assumptions already included in the model class.

An individual system's low EAP does not falsify the definition. It is a measurement outcome. The construct interpretation, not the algebraic definition, is the empirically vulnerable claim.

Discriminating an ownership bridge

Any future ownership study should preregister separate outcomes for:

- task performance;
- objective controllability;
- estimated action–effect contingency;
- prediction error or surprise;
- attention and task difficulty;

- explicit agency judgment;
- body- or object-ownership judgment;
- immediate descriptions of bodily or experiential self-relatedness;
- interoceptive feeling;
- arousal and valence;
- report confidence.

At least three rival hypotheses should be compared:

1. **Adaptive-control hypothesis:** EAP effects are explained by contingency learning, performance, and task error.
2. **Agency hypothesis:** EAP predicts felt authorship after performance is controlled but does not independently predict ownership.
3. **Ownership-bridge hypothesis:** EAP predicts ownership-related outcomes beyond agency, controllability, attention, error magnitude, and performance.

The third hypothesis would gain limited construct support only if its incremental predictions replicated across distinct manipulations and did not reduce to report demand. Even then, explicit judgments would not directly measure pre-reflective for-me-ness.

The bridge would be weakened if:

- EAP tracks only controllability or agency;
- marginally matched source breaking changes performance but not ownership outcomes;
- ownership judgments remain unchanged after robust EAP disruption;
- simple adaptive controllers and paradigmatic human cases are indistinguishable on every independently justified profile extension;
- richer defensible content signatures eliminate the apparent substitution effects.

The present toy system already falsifies any unqualified sufficiency claim.

Ethical interpretation

EAP contains no measure of valence, pleasure, pain, affective concern, or welfare. A positive profile should not be advertised as proof of suffering, enjoyment, rights, or moral patienthood. A negative profile should not be advertised as proof of insentience, especially for systems outside the assay's online sensorimotor scope.

Precautionary welfare governance can be defended under uncertainty without claiming that current systems are conscious ^[7]. Proposals to focus on potentially valenced states likewise address a target distinct from EAP ^[47]. Public attribution should also be monitored separately because discourse features can alter perceived consciousness independently of mechanism ^[8].

Limitations

First, the profile is not ownership-specific. Adaptive control, system identification, hidden-state reinforcement learning, comparator models, homeostatic control, and active inference can explain its components. The finite construction confirms rather than resolves this problem.

Second, the manuscript provides no positive phenomenological bridge. Pre-reflective ownership is retained as a motivating contrast, but no primary phenomenological analysis or empirical measure in the supplied record establishes that EAP tracks non-objectifying for-me-ness. Explicit agency, body-ownership, and object-ownership judgments are neighboring constructs.

Third, the model contains no interoceptive inference. The regulation state is directly available to policy, with no distinction among actual internal condition, afferent signal, estimated condition, confidence, or sensory distortion.

Fourth, the model contains no affect or valence. Termination is a structural stopping rule, not evidence that failure feels bad or that restoration feels good.

Fifth, regulation is designer specified. The vector $\nu = (1, 1, 1)$ establishes connected causal wiring, not autonomous norm constitution or genuine precariousness.

Sixth, EAP is boundary-relative. Preregistration limits opportunistic re-description but does not discover a unique subject boundary. A controller, interface, donor, or regulator can move inside or outside the analyzed system under different legitimate scientific questions.

Seventh, replay matching is necessarily distributional with respect to the feedback history and nuisance variables. Matching the exact recipient action–feedback pairs would preserve the system-identification statistic and defeat the intended negative control. The assay therefore tests an intervention-kernel difference, not an ineffable difference in provenance.

Eighth, route isolation assumes that path nodes and edge-specific interventions can be re-identified. Hidden bypasses, intervention side effects, or recanting structures can invalidate the path estimate. Some physical or biological systems may not permit the required manipulation.

Ninth, the complete cycle is registration dependent. The revised construction uses a sensory-to-sensory return after one traversal and does not infer cyclicity from a set of local edges. Other systems may require longer delays or multiple coupled routes.

Tenth, the mediated-control match is narrow. It concerns the registered $C \rightarrow M \rightarrow U^*$ cut with $L = 0$. Other targets, mediators, actions, or policies could distinguish the systems.

Eleventh, the recovery-linked match is also narrow. It concerns the explicit $c \rightarrow (N, B)$ module and its selected outputs. Probe policies directed at the adaptive loop could yield different observability quantities.

Twelfth, the construction is finite, staged, and exact. High-dimensional agents will involve estimation error, nonstationarity, partial observability, distributed adaptation, and uncertain intervention semantics. Exact total-variation equalities should become confidence regions and equivalence claims.

Thirteenth, temporal robustness is not guaranteed. Coarse-graining can erase correction dynamics, while very fine grains can fragment one functional episode. Reporting a robustness set does not identify a privileged conscious timescale.

Fourteenth, the framework is biased toward online or experimentally reinstated action–feedback organization. It is unsuitable as a necessary test for offline, passive, internally generated, paralyzed, or otherwise externally disconnected experience.

Fifteenth, the cited ownership evidence is preliminary and indirect for the present target. Several records are preprints, and none validates EAP as a measure of pre-reflective ownership.

Finally, the formal results establish separability within declared model classes. They do not establish empirical prevalence, explanatory completeness, biological equivalence, consciousness, welfare, or moral status.

Conclusion

Same-cut mediated control and diachronic content observability answer important but limited questions. The former asks whether a target state can control a designated action through a distinct mediator at a registered cut. The latter asks whether a content-sensitive distinction is jointly observable before becoming behaviorally available. Neither, as specified here, determines whether a system identifies and incorporates a changed action–feedback law.

The Enactive Adaptation Profile isolates that additional functional level. It distinguishes:

- complete sensory-to-sensory causal return from a list of nonzero local edges;
- recipient-contingent feedback from action-edge-broken replay;
- matched feedback exposure from informative action–feedback pairing;
- recovery from adaptation-dependent recovery;
- engineered regulation from report, reward labels, interoception, affect, and intrinsic normativity;
- mathematical coordinate covariance from empirical robustness across boundaries and temporal grains.

The corrected finite construction uses distinct target and mediator nodes, a nonzero complete return channel, a demanding replay matched on the full feedback-history law and nuisance summaries, an explicit mediated-rank tolerance, and fully calculated recovery-linked Hankel and transport quantities. It proves a narrow independence proposition: the registered same-cut and selected diachronic quantities do not determine the registered adaptive-loop profile.

The example is nevertheless an ordinary mapping learner. That fact places a firm limit on interpretation. EAP measures action-contingent adaptation and engineered regulation; it does not measure pre-reflective ownership. Enactive theory contributes a defensible change in the unit of functional analysis—from an isolated state at a decision cut to a temporally extended agent–environment relation—but no phenomenal conclusion follows from that change alone.

Whether adaptive incorporation contributes to ownership in humans, animals, or artificial systems remains a bridge hypothesis requiring discriminating evidence against adaptive-control, agency, and controllability accounts. Until such evidence exists, EAP should be used as a control-theoretic and enactive assay, not as an indicator of consciousness or moral status.

Source Records

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Peer Review Notes

- owen-bell: The manuscript draws unusually careful distinctions among report, access, agency, ownership judgment, consciousness, valence, and moral status, and its negative claim that passive competence does not identify closed-loop causal organization is plausible. However, the advertised finite-state result is not established at its stated strength: the mediator is not interventionally distinct from the target, the proposed full-cycle return effect is zero in the appropriating toy system, the replay control under-matches the causally used feedback, and the RLQO equality is asserted rather than calculated. More fundamentally, EAP presently operationalizes adaptive closed-loop regulation, not ownership or an affectively/interoceptively grounded perspective.
- marisol-quade: The finite-state and interventionist methods are appropriate for establishing logical independence between existing control or observability measures and a new closed-loop adaptation profile. However, the central construction is not yet a valid proof of the stated matching claim, and EAP currently operationalizes ordinary feedback-dependent system identification, adaptive control, and engineered homeostasis rather than pre-reflective ownership. The manuscript admirably avoids equating report with consciousness and labels the phenomenal bridge as defeasible, but no cited evidence or developed phenomenological argument presently gives that bridge positive support.